Actividad
Divergencia y Rotacional

1). Calculo de Divergencia y 3 puntos en el plano

$f(x,y) = (x^3, -y^3)$ / $(1,2), (3,1), (4,4)$

$\nabla \cdot f(x,y) = g(x,y) = 3x^2 - 3y^2$ / $g(1,2) = -9$, $g(3,1) = 24$, $g(4,4) = 0$

entra más sale menos — sale más entra menos — sale igual que entra

2). $f(x,y) = (-x^3, y^3)$ / $(1,1), (-1,-1), (5,2)$

$\nabla \cdot f(x,y) = -3x^2 + 3y^2 = g(x,y)$ / $g(1,1) = 0$, $g(-1,-1) = 0$, $g(5,2) = -63$

entra más
sale más
entra = sale

3). $f(x,y) = (x^2y^2, \sin(xy))$ / $(0,1), (1,0), (\pi,0)$

$\nabla \cdot f(x,y) = g(x,y) = 2xy^2 + \cos(xy) \cdot x$ / $g(0,1) = 0$, $g(1,0) = 1$, $g(\pi,0) = \pi$

4). $f(x,y) = (\sin(x)\cos(y), -\cos(y^2))$ / $(0,\pi), (\pi, \frac{\pi}{2}), (\frac{3\pi}{2}, -\pi)$

$\nabla \cdot f(x,y) = g(x,y) = \cos(x)\cos(y) + 2\sin(y^2) \cdot y$ / $g(0,\pi) = -1 + 2\sin(\pi^2)\pi < 0$

$g(\pi, \frac{\pi}{2}) = 0 + \pi \sin(\frac{\pi^2}{4}) > 0$

$g(\frac{3\pi}{2}, -\pi) = 0 + (-2\pi \sin(\pi^2) > 0$

5). $f(x,y) = (x^3 + 4x^2y, -5xy^2 + y^3)$ / $(0,1)$ $(1,2)$ $(1,-1)$

$\nabla \cdot f(x,y) = g(x,y) = 3x^2 + 8xy - 10xy - 3y^2$ / $g(0,1) = -3$, $g(1,2) = -13$

$= 3x^2 - 2xy - 3y^2$ $\quad g(1,-1) = 2$

6).

$f(x,y) = (\cos(x^3) + 4\sin(x^2)\cos y, -5x^2y^3 + 2x^3y^2)$

$(0,0), (\pi,0), (0,\pi)$

$\nabla \cdot f(x,y) = g(x,y) = -\sin(x^3)3x^2 + 8x\cos(x^2)\cos y - 15x^2y^2 + 4x^3y$

$g(0,0) = 0 + 0 - 0 + 0 = 0$, $g(\pi,0) = -\sin(\pi^3)3\pi^2 + 8\pi\cos(\pi^2) < 0$

$g(0,\pi) = 0 + 0 - 0 + 0 = 0$

Rotacional

1). $\nabla \times f(x,y) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^3 & -y^3 & 0 \end{vmatrix}$ Coofactores $= g(x,y)$

$f(x,y) = (x^3, -y^3)$ $\quad g(x,y) = \hat{i}\left(\frac{\partial 0}{\partial y} - \frac{\partial -y^3}{\partial z}\right) - \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial z}\right)\hat{j} + \hat{k}\left(\frac{\partial -y^3}{\partial x} - \frac{\partial x^3}{\partial y}\right)$

$(1,2), (3,1), (4,4)$ / $= (0,0,0) \rightarrow$ Conservativa

$g(1,2) = g(3,1) = g(4,4) = (0,0,0)$

2). $f(x,y) = (-x^3, y^3)$ / $(1,1)$ $(-1,-1)$, $(5,2)$

$\nabla \times f(x,y) = g(x,y) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -x^3 & y^3 & 0 \end{vmatrix} = \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(0-0)$

Conservativo

$= (0,0,0) = g(1,1) = g(-1,-1) = g(5,2)$

3). $f(x,y) = (x^2y^2, \sin(xy))$ / $(0,1)$, $(1,0)$, $(\pi,0)$

$\nabla \times f(x,y) = g(x,y) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y^2 & \sin(xy) & 0 \end{vmatrix} = \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(\cos(xy)y - 2x^2y)$

- Rotacion Antihoraria
- Rotacion horaria
- No rota

$= (0, 0, y\cos(xy) - 2x^2y)$

$g(0,1) = (0,0,1) \rightarrow g(1,0) = (0,0,0)$, $g(\pi,0) = (0,0,0)$

4) $f(x,y) = (\sin(x)\cos(y), -\cos(y^2))$ / $(0,\pi), (\pi, \frac{\pi}{2}), (\frac{3\pi}{2}, -\pi)$

$$\nabla \times f(x,y) = g(x,y) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin(x)\cos(y) & -\cos(y^2) & 0 \end{vmatrix} = \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(0 + \sin(x)\sin(y))$$

$$= (0, 0, \sin(y)\sin(x))$$

$g(0,\pi) = (0,0,0)$

$g(\pi, \frac{\pi}{2}) = (0,0,0)$

$g(\frac{3\pi}{2}, -\pi) = (0,0,0)$

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5). $f(x,y) = (x^3 + 4x^2y, -5xy^2 - y^3)$ / $(0,1), (1,2), (1,-1)$

$$\nabla \times f(x,y) = g(x,y) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^3 + 4x^2y & -5xy^2 - y^3 & 0 \end{vmatrix} = \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(-5y^2 - 4x^2)$$

$$= (0, 0, -5y^2 - 4x^2)$$

$g(0,1) = (0,0,-5)$

$g(1,-1) = (0,0,-9)$

$g(1,2) = (0,0,-24)$

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6). $f(x,y) = (\cos(x^3) + 4\sin(x^2)\cos y, -5x^2y^3 + 2x^3y^2)$

$(0,0), (\pi,0), (0,\pi)$

$$\nabla \times f(x,y) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \cos x^3 + 4\sin x^2 \cos y & -5x^2y^3 + 2x^3y^2 & 0 \end{vmatrix} = \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}\left(\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial x}\right)$$

$$= (0, 0, +4\sin(x^2)\sin y - (10xy^3 + 6x^2y^2)$$

$$= (0, 0, 4\sin(x^2)\sin y - 10xy^3 + 6x^2y^2)$$

$g(0,0) = (0,0,0)$

$g(\pi,0) = (0,0,0)$

$g(0,\pi) = (0,0,0)$